

Technical Brief: Varsity Cyber Clash Network Infrastructure Design

1. Project Overview and Design Objectives

The Varsity Cyber Clash is a high-profile regional E-Sports tournament hosted at the Universiti Teknikal Malaysia Melaka (UTeM) campus. As a mission-critical event organized by the University Student Council, the network must support a highly competitive environment where millisecond latency can determine the outcome of professional-level match play.

The core architectural challenge is to deliver a robust, high-performance network while strictly adhering to a "minimal cost" constraint. The designer is tasked with balancing expensive high-bandwidth requirements for broadcasting and gaming with the fiscal necessity of utilizing cost-effective hardware, ensuring every component selection is technically justified and provides the highest possible ROI.

2. Physical Tiered Architecture

The network must adhere to a Three-Tier Hierarchical Star topology to ensure modularity and ease of troubleshooting:

- Core Router: Serves as the authoritative gateway for external internet connectivity. It manages inter-VLAN routing, providing the necessary logical pathing between user zones and the central server farm.**
- Core Switch: Must utilize L3 Gigabit switching. Per the D-Link enterprise categorization, this device is required for its support of dynamic routing protocols (RIP/OSPF) and its**

capability to act as the central routing engine for a large IP routed network, aggregating traffic from all departmental-level switches.

- **Access Tier: Deployed at the network edge using 48-port Managed L2 Switches. Managed switches are mandatory at this tier—over unmanaged or smart alternatives—to support essential VLAN tagging, Quality of Service (QoS) for gaming traffic, and port-level security for 70+ wired endpoints.**

3. User Zone Specifications and Connectivity

The following table defines the connectivity requirements for the three primary user groups based on the anticipated event density:

User Zone	Connection Type	User Count	Hardware/Medium
Wired Players	Wired	70	Dedicated CAT6 Cabling
Wired Admins/Broadcasters	Wired	30	Dedicated CAT6 Cabling
Wireless Spectators	Wireless	300	6 x Wi-Fi 6 Access Points

4. Server Farm and Service Infrastructure

The designer must implement a dedicated server zone. Services are categorized into mandatory event operations and advanced performance enhancements:

Mandatory Services

- **Game Servers:** Hosting dedicated match instances for low-latency play.
- **Web & Registration:** Managing participant data and public event information.
- **FTP (File Transfer Protocol):** Facilitating resource sharing and tournament file distribution.
- **Live Streaming Platform:** Encoding and broadcasting high-definition feeds.
- **Tournament Management System:** Real-time bracketing and score coordination.

Advanced Services

- **VoIP (Voice over IP):** Providing dedicated voice communication channels for teams.
- **Caching:** Reducing repetitive data requests to decrease overall latency.
- **Active Directory:** Centralized management of user authentication and network resources.

5. Logical Mapping: VLAN and IP Strategy

To isolate broadcast domains and enhance security, the network shall be segmented as follows:

- 1. VLAN 10: Players**
- 2. VLAN 20: Admins**

3. VLAN 30: Servers

4. VLAN 40: Spectators

5. VLAN 99: Management

IP Scheme Design Instruction: Develop a private IPv4 addressing scheme using CIDR notation. Note that the spectator zone (VLAN 40) requires a minimum of a /23 subnet (e.g., 172.16.0.0/23) to accommodate the 300-user count while leaving sufficient headroom for guest growth. All other subnets should be sized according to their specific host requirements plus a 20% growth margin.

6. Technical Performance and Signal Integrity

Transmission Impairment Mitigation Designers must account for attenuation and noise in the high-density UTeM environment. To preserve signal integrity, CAT6 cable runs must not exceed 100 meters; for longer distances, repeaters or amplifiers must be used to restore signal energy. Specific shielding strategies must be implemented to mitigate crosstalk and thermal noise, which can degrade the signal-to-noise ratio in high-density E-sports setups.

Throughput & Bandwidth Optimization The design must respect the theoretical limits of the medium. Utilize the Shannon Capacity formula, $C = B \log_2(1 + \text{SNR})$, to ensure that the bandwidth (B) allocated to gaming and streaming channels is sufficient to handle peak data rates without reaching the channel capacity limit. Digital signals must be optimized with

high bit rates and minimal bit intervals to ensure maximum throughput for real-time data flow.

Digital Data Multiplexing Synchronous Time-Division

Multiplexing (TDM) principles must be applied to all core links.

The core link data rate must be n times faster than individual access lines, where n represents the number of input sources.

Consequently, the unit duration on the multiplexed link must be n times shorter than the duration of the source data units to ensure seamless interleaving.

7. Implementation and Troubleshooting Requirements

The following verification steps are mandatory to validate the network deployment:

- **[] System Information Validation: Document Processor, RAM, and OS on all gaming workstations to ensure hardware baseline compliance.**
- **[] Configuration Audit: Execute `ipconfig /all` on representative hosts in each VLAN to verify correct DNS, Gateway, and Subnet Mask assignments.**
- **[] Connectivity Verification: Use ping to validate successful paths from User Zones to the Server Farm and identify any firewall-induced latency.**
- **[] Resource Sharing Test: Verify accessibility of shared network drives, folders, and removable drives across the FTP zone.**

- [] **Peripheral Access: Confirm shared resource functionality for network printers, following standard sharing procedures.**

8. Cost Optimization Constraints

Architect's Note: Every equipment selection must be strictly justified against the "Minimal Cost" mandate. You must utilize L2 Gigabit Managed switches at the access tier for all endpoint connectivity; L3 Gigabit equipment is reserved exclusively for the core where dynamic routing (RIP/OSPF) and inter-departmental aggregation are technically required. Reference the D-Link enterprise categorization to ensure we are not over-provisioning at the edge while maintaining the high-capacity backbone required for the UTeM regional tournament.